

Project Risks and Risk Management

ID	Risk Description	Possible Cause	Probability	Impact	Mitigation / Response Strategy
R1	Requirements may change or remain unclear	Pharmacy procedures may not initially be completely understood	Medium	High	Meet with the pharmacy stakeholder regularly, document requirements clearly, and use incremental development so changes can be incorporated between increments
R2	Prescription information or customer information may be accessed by unauthorized users	Incorrect access control or insecure system design	Medium	High	Implement authentication, role-based permissions, secure password handling, and restrict prescription access to authorized staff
R3	Integration between different system components may fail	Prescription, ordering, payment, and inventory modules depend on each other	Medium	High	Define interfaces and database structure early, integrate incrementally, and perform integration testing after each increment
R4	The team may fall behind schedule	University workload, task complexity, or poor task distribution	Medium	High	Divide work into small milestones, assign responsibilities early, track progress through GitHub, and prioritize core features
R5	Team members may have insufficient experience with required technologies	New frameworks, databases, authentication, or web-development tools	Medium	Medium	Select familiar technologies when possible, learn required technologies early, divide tasks based on skills, and build small prototypes

R6	Inventory data may become inconsistent	Incorrect updates, cancelled orders, duplicate operations, or integration errors	Medium	High	Centralize inventory updates, validate quantities before checkout, test inventory transactions, and prevent stock values from becoming negative
R7	Payment functionality may be difficult to implement safely	Complexity of real payment-gateway integration	Medium	High	Use a test/sandbox or simulated payment process for the academic prototype and separate payment processing from core business logic
R8	Pharmacy stakeholders may not be available for continuous feedback	Owner/staff workload and limited meeting availability	Medium	Medium	Prepare questions before meetings, document decisions, request feedback at major milestones, and avoid unnecessary meetings

The lecture emphasizes that requirements should be **realistic and verifiable**, rather than vague or impossible to test.